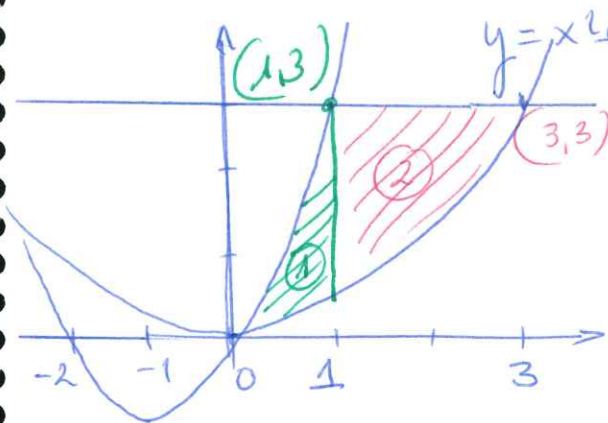


Et OHIKOA - 2024

$$\begin{aligned} y &= x^2/3 \\ y &= x^2+2x \\ y &= 3 \end{aligned}$$

1) Marraztu 1. koadrantean mugatutako azalera.

$$y = \frac{x^2}{3} \rightarrow \text{Erpina } y' = 2x/3 \\ 2x/3 = 0 \rightarrow x = 0 \rightarrow y = 0 \in (0,0)$$



$$y = x^2 + 2x \rightarrow y' = 2x + 2 \\ 2x + 2 = 0 \rightarrow x = -1 \rightarrow y = -1 \in (-1,-1)$$

Ardatutik

$$\begin{aligned} y &= 0 \\ y &= x^2 + 2x \end{aligned} \quad \begin{aligned} x^2 + 2x &= 0 \\ x(x+2) &= 0 \end{aligned} \quad \begin{aligned} x &= 0 \\ x &= -2 \end{aligned}$$

Erabakitze puntuak kalkulatzeko:

$$\begin{aligned} \begin{cases} y &= x^2/3 \\ y &= x^2+2x \end{cases} & \quad \begin{aligned} \frac{x^2}{3} &= x^2+2x \\ x^2 &= 3x^2+6x \\ -2x^2-6x &= 0 \end{aligned} & \quad \begin{aligned} -2x(x+3) &= 0 \\ x &= 0 \\ x &= -3 \end{aligned} \end{aligned}$$

~~$x = -3$~~ eta 1. koadrantean

$$\begin{aligned} \begin{cases} y &= x^2/3 \\ y &= 3 \end{cases} & \quad \begin{aligned} 3 &= x^2/3 \\ 9 &= x^2 \end{aligned} & \quad \begin{aligned} x &= 3 \rightarrow (3,3) \\ x &= -3 \end{aligned} \end{aligned}$$

~~$x = -3$~~ eta 1. koadr.

$$\begin{aligned} \begin{cases} y &= x^2+2x \\ y &= 3 \end{cases} & \quad \begin{aligned} 3 &= x^2+2x \\ x^2+2x-3 &= 0 \end{aligned} & \quad \begin{aligned} x &= 3 \\ x &= -3 \end{aligned} \end{aligned}$$

~~$x = 3$~~ eta 1. koadrantean. $x = -3 \rightarrow (-3, -3)$

Azalera bi esparrutan banatzen dugu:

$$A = \int_0^1 (x^2+2x - \frac{x^2}{3}) dx + \int_1^3 (3 - \frac{x^2}{3}) dx =$$

$$\int_0^1 (\frac{2x^2}{3} + 2x) dx + \int_1^3 (3 - \frac{x^2}{3}) dx$$

$$A = \left[\frac{2x^3}{9} + \frac{2x^2}{2} \right]_0^1 + \left[3x - \frac{x^3}{9} \right]_1^3$$

Barrower eregek
aplikator:

$$= \left(\frac{2}{9} + 1 \right) + (9 - 3) - \left(3 - \frac{1}{9} \right) = \frac{7}{6} + 6 - \frac{26}{9} = \frac{13}{3} u^2$$